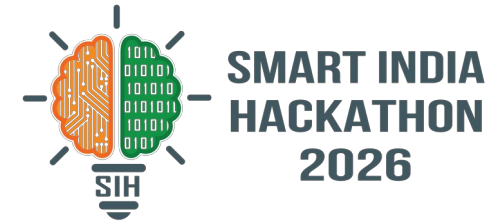


SMART INDIA HACKATHON 2026



FleetMesh

**Local coordination for
warehouse robot fleets**

Problem Statement ID: 26123

Edge-AI Based Distributed Fleet Coordination for
Autonomous Mobile Robots (AMRs) in Smart
Warehouses

Theme: Smart Automation

PS Category: Software

Team ID: [Enter registered ID]

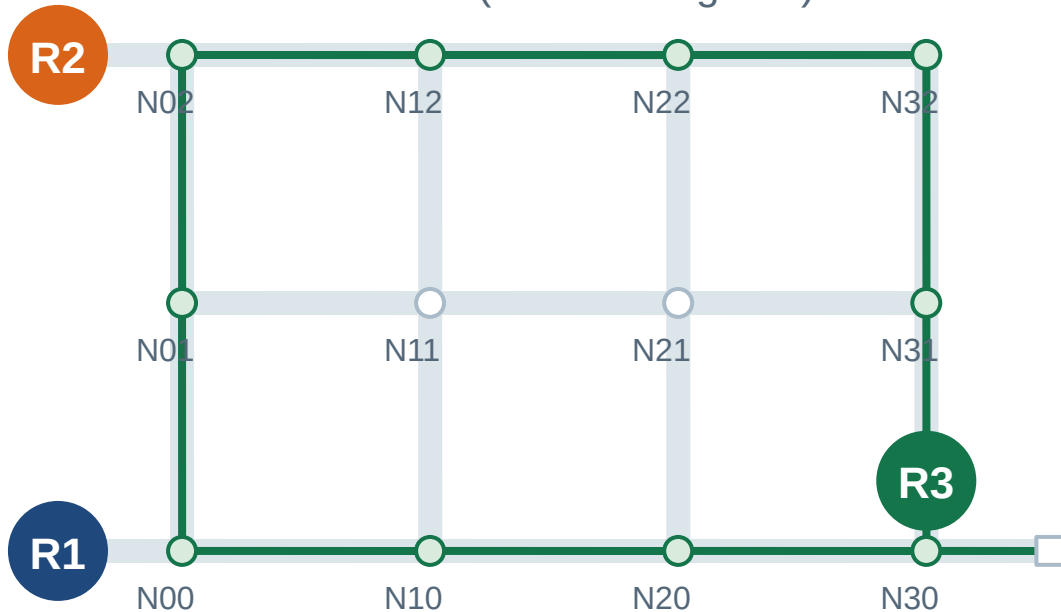
Team Name: [Enter registered team name]



Proposed Solution

Robots agree before shared routes overlap

Simulator state at 20.0 s (fixed routing rule)



R1 and R2 wait for grants. R3 is moving.

Detailed explanation of the proposed solution

Each robot owns its planner.
Peers share intent and grant routes.

How it addresses the problem

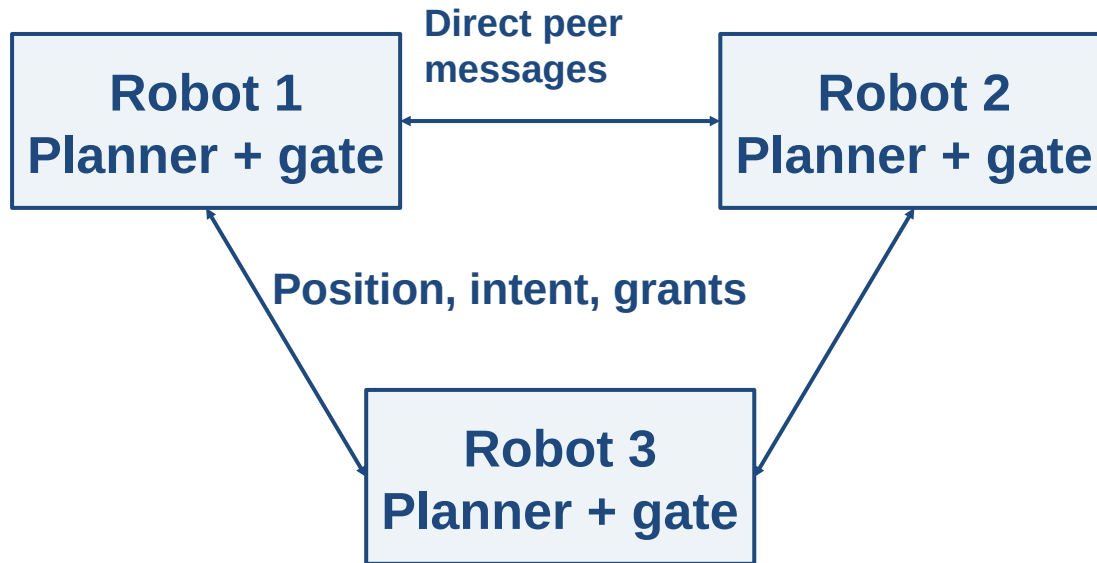
Release cleared zones early.
Replan or wait at blocked aisles.

Innovation and uniqueness of the solution

Inspect decisions. Replay and review.

Methodology and process for implementation

Each robot plans locally. Peers agree on access.



One laptop hosts the independent processes.

Technologies to be used

Python processes + authenticated UDP
SQLite + HTML/CSS/JS console

01 Bid for jobs

Agree on pickup ownership.

02 Plan the route

Local regression or fixed route costs.

03 Reserve and move

Peer grants, then release clear zones.

Analysis of the feasibility of the idea

3–6

independent robot processes

160

completed benchmark runs

0

detected collisions in those runs

Potential challenges and risks	Strategies for overcoming these challenges
Failed peer can halt progress	Now: retain owned zones. Next: verified restart recovery.
Idealized sensing and motion	Next: edge-board tests, braking trials and noisy sensing.
Predictor trails the fixed rule	Now: fixed rule by default. Next: independent AI tests.

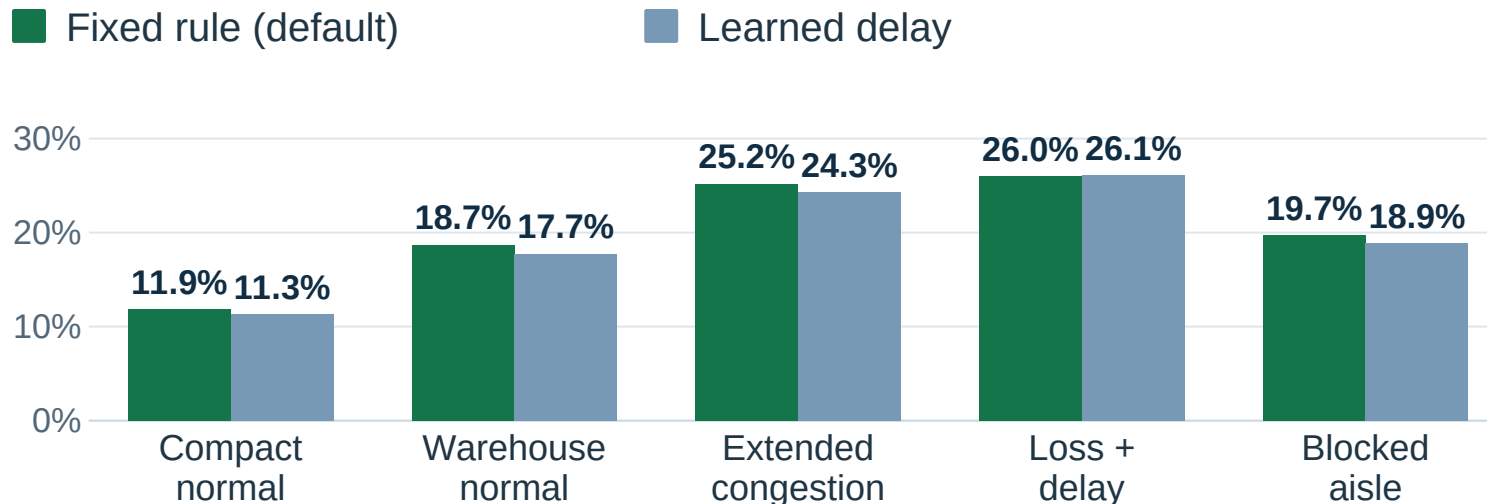
26 system tests. 6 additional process cases: 81 jobs completed without collisions.

Potential impact on the target audience

For AMR manufacturers and warehouse integrators

Pilot offer: one warehouse map, then three AMRs. Revenue: integration + support.

Benefits of the solution



25.2%

less time in extended
congestion

26.0%

less time with loss and delay

Both vs the included sequential baseline. 8 runs per bar. Return to dock included.
20% target met in 2 of 5 means. The fixed rule is faster in 4 of 5 scenario averages.

Details / Links of the reference and research work

01 Problem statement

SIH 2026, PS 26123
Requirements from the supplied screenshots.

02 Coordination method

Ricart–Agrawala (1981)
Peer mutual exclusion as the foundation.

03 Safety context

Nav2 Collision Monitor
Physical safety needs hardware validation.

04 Existing fleet products

MiR Fleet and [OTTO](#)
Fleet coordination already exists commercially.

05 Open coordination research

Open-RMF traffic management (2026)
A relevant open-source comparison.

06 Our evidence package

160 benchmark runs + 26 system tests
34 UI/API checks + recorded run review.

Demo sequence Urgent job. Closed aisle. Message outage. Recorded recovery.